

Extended sign-kill for $\widehat{\lambda} = (5, 2)$: four-piece E_{n-1}^+ decomposition, two hook branches

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Abstract

Let $\lambda = (5, 2, 1^q) \vdash n = q + 7$ with $q \geq 5$, $\ell = q + 2 = n - 5$, and $B = B_{\ell+1}^{(\lambda)} V_\lambda = R'_{\ell+1} R'_{\ell+2} R'_{\ell+3} R'_{\ell+4} R'_{\ell+5} V_\lambda$ (five R' -blocks). We prove the “ $\subseteq$ ” direction of the extended sign-kill conjecture for this family at the predicted threshold $\tau = q + 3 - \widehat{\lambda}_1 - \widehat{\lambda}_2 = q - 4$:

$$B \subseteq \bigcap_{j=1}^{q-4} E_j^- |_{V_\lambda}.$$

At $q \in \{0, 1, 2, 3, 4\}$ the predicted threshold is non-positive, so the containment is vacuous.

The shape $\widehat{\lambda} = (5, 2)$ has $r = 2$ length-preserving corners, so the inner E_{n-1}^+ decomposition has four pieces (three corner-pair Φ -pieces and one same-row Φ_h -piece) — the same total count as the eighth-pass $\widehat{\lambda} = (4, 3)$ proof. The new combinatorial feature: *two* of the four pieces are hooks (vs. one hook in $(4, 3)$). Specifically, the contributing branches are $\mu_A = (4, 2, 1^{q-1})$ (non-hook, $\widehat{\lambda} = (4, 2)$) and $\mu_B = (5, 1^q)$ (hook), and the vanishing branches are $\mu_C = (4, 1^{q+1})$ (hook) and $\mu_h = (3, 2, 1^q)$ (non-hook, $\widehat{\lambda} = (3, 2)$). The prefix bound on the μ_B piece comes from the May-14 fifth-pass hook support lemma at $k = 5$, matching $q - 3$ to the recursive $(4, 2)$ prefix bound on the μ_A piece. Dimensions match: $\dim B = f^{(4,1)} = 4 = f^{(3,1)} + f^{(4)} = \dim B_{\ell+1}^{(\mu_A)} + \dim B_{\ell+1}^{(\mu_B)}$.

This completes the $\widehat{\lambda}_1 + \widehat{\lambda}_2 = 7$ row of the recursive ladder.

1 Setup

We use the conventions of the May-13–May-14 series. $H_q(S_n)$ is the Iwahori–Hecke algebra over $\mathbb{Q}(q)$ with generators $T_1, \dots, T_{n-1}$ and quadratic relation $T_j^2 = (q - 1)T_j + q$. V_λ is the Specht module in the Hoefsmit seminormal basis $\{v_T : T \in \text{SYT}(\lambda)\}$.

For $1 \leq j \leq n - 1$, set $S_j := T_j + 1$. Write

$$E_j^+ := \ker(T_j - q) |_{V_\lambda} = \text{Im}(S_j |_{V_\lambda}), \quad E_j^- := \ker(S_j) |_{V_\lambda}.$$

For $k \geq 1$, let $R'_k := S_{k-1} S_{k-2} \cdots S_1$ with the convention $R'_1 = 1$. For $\lambda \vdash n$ with $\ell = \ell(\lambda)$, set

$$\Omega = \Omega^{(\lambda)} := R'_{\ell+1} R'_{\ell+2} \cdots R'_n, \quad B := \Omega \cdot V_\lambda = B_{\ell+1}^{(\lambda)} V_\lambda.$$

For $\lambda = (5, 2, 1^q)$, $n = q + 7$ and $\ell = q + 2$, so $\Omega = R'_{q+3} R'_{q+4} R'_{q+5} R'_{q+6} R'_{q+7} = R'_{n-4} R'_{n-3} R'_{n-2} R'_{n-1} R'_n$ has *five* R' -blocks. By the May-13 multi-corner-dimension paper, $\dim B = f^{\lambda^{(\geq 2)}} = f^{(4,1)} = 4$ in the stable regime $q \geq 2$.

Removable corners of λ . The removable corners are

$$c_1 = (1, 5), \quad c_2 = (2, 2), \quad c_3 = (q + 2, 1).$$

λ has two length-preserving corners (c_1, c_2) and one column-1 bottom corner (c_3) , an $r = 2$ shape.

Identity prefix and seminormal support. For an SYT T of any partition μ , the *identity prefix length* is

$$p(T) := \max\{P \geq 0 : T(i, 1) = i \text{ for all } 1 \leq i \leq P\}.$$

For $W \subseteq V_\mu$, $\text{supp}(W) := \{T \in \text{SYT}(\mu) : \exists w \in W, [v_T]w \neq 0\}$.

2 Input from prior papers

Proposition 1 (Support rule, May-14 fourth pass). *Let $W \subseteq V_\nu$ be any subspace, $1 \leq j \leq |\nu| - 1$. If every $T \in \text{supp}(W)$ has j and $j + 1$ in the same column of T , then $W \subseteq E_j^-|_{V_\nu}$.*

Remark 2. If every $T \in \text{supp}(W)$ has $p(T) \geq P$, then for every $1 \leq j \leq P - 1$ the entries $j, j + 1$ sit at column-1 rows $j, j + 1$, sharing column 1. By Proposition 1, $W \subseteq E_j^-$ for $1 \leq j \leq P - 1$.

Theorem 3 (Extended sign-kill at $\hat{\lambda} = (3, 2)$, May-14 fifth pass). *For $\lambda' = (3, 2, 1^{q'}) \vdash n' = q' + 5$ with $q' \geq 3$, every $T' \in \text{supp}(B_{q'+3}^{(\lambda')} V_{\lambda'})$ satisfies $p(T') \geq n' - 6 = q' - 1$. Consequently, $B_{q'+3}^{(\lambda')} V_{\lambda'} \subseteq E_j^-|_{V_{\lambda'}}$ for $1 \leq j \leq q' - 2$, and in particular $B_{q'+3}^{(\lambda')} \subseteq E_1^-$ for $q' \geq 3$.*

Theorem 4 (Extended sign-kill at $\hat{\lambda} = (4, 2)$, May-14 sixth pass). *For $\lambda' = (4, 2, 1^{q'}) \vdash n' = q' + 6$ with $q' \geq 4$, every $T' \in \text{supp}(B_{q'+3}^{(\lambda')} V_{\lambda'})$ satisfies $p(T') \geq n' - 8 = q' - 2$. Consequently, $B_{q'+3}^{(\lambda')} V_{\lambda'} \subseteq E_j^-|_{V_{\lambda'}}$ for $1 \leq j \leq q' - 3$, and in particular $B_{q'+3}^{(\lambda')} \subseteq E_1^-$ for $q' \geq 4$.*

Remark 5. Theorem 4 as stated in the sixth-pass paper has a mild gap (its $E_{n'-1}^+$ decomposition silently omits the same-row $+q$ -piece at $c_1 = (1, 4)$ with inner shape $(2, 2, 1^{q'})$). The eighth-pass paper (Remark 3.7 there) and the present paper note that this gap is closable by adding an extra branch handled by the $\hat{\lambda} = (2, 2)$ extended sign-kill theorem (May-14 fourth pass). The conclusion of Theorem 4 (containment direction) is correct as stated; we apply it below.

Theorem 6 (Hook column-1, May-12). *For $\mu = (k, 1^m) \vdash n_0$ with $k \geq 2$ and $m \geq \max(k, 2)$, $B_{\ell_0+1}^{(\mu)} V_\mu \subseteq E_1^-|_{V_\mu}$ (where $\ell_0 = \ell(\mu) = m + 1$).*

Theorem 7 (Hook support lemma, May-14 fifth pass). *Let $\mu = (k, 1^m) \vdash n_0 := k + m$ with $k \geq 2$ and $m \geq k$. Set $\ell_0 := \ell(\mu) = m + 1$ and $j_0 := \ell_0 + 1 = m + 2$. Then every $T \in \text{supp}(B_{j_0}^{(\mu)} V_\mu)$ satisfies $p(T) \geq m - k + 2$.*

Theorem 8 (Phase A endpoint, May-19). *For any H_q -module V and any $1 \leq j \leq n - 1$, $(T_j + 1) \cdots (T_1 + 1)V = E_j^+(V)$. In particular, $R'_n V_\lambda = E_{n-1}^+|_{V_\lambda}$.*

Lemma 9 (Outer-factor preservation, May-14 fifth pass). *Let $W \subseteq V_\nu$ be a subspace such that every $T \in \text{supp}(W)$ satisfies $p(T) \geq P$. Then for every $j \geq P + 1$, every $T'' \in \text{supp}((T_j + 1)W)$ also satisfies $p(T'') \geq P$.*

3 The four-piece E_{n-1}^+ decomposition for $(5, 2, 1^q)$

The $+q$ -eigenspace $E_{n-1}^+|_{V_\lambda}$ admits a clean four-piece decomposition reflecting the corner structure of λ .

Corner-pair pieces. For each pair $\{c_X, c_Y\}$ of distinct removable corners, placement of $\{n-1, n\}$ at $\{c_X, c_Y\}$ in either order yields a T_{n-1} -2-block in V_λ . The pairs and axial distances:

- $\{c_1, c_2\} = \{(1, 5), (2, 2)\}$. Content $c(1, 5) - c(2, 2) = 4 - 0 = 4$. Non-degenerate ($|d| = 4 \geq 2$). Inner shape $\mu_C := (4, 1^{q+1})$.
- $\{c_1, c_3\} = \{(1, 5), (q+2, 1)\}$. Content $c(1, 5) - c(q+2, 1) = 4 - (1 - (q+1)) = q+4$. Non-degenerate ($|d| \geq 9$ at $q \geq 5$). Inner shape $\mu_A := (4, 2, 1^{q-1})$.
- $\{c_2, c_3\} = \{(2, 2), (q+2, 1)\}$. Content $c(2, 2) - c(q+2, 1) = 0 - (1 - (q+1)) = q$. Non-degenerate ($|d| \geq 5$ at $q \geq 5$). Inner shape $\mu_B := (5, 1^q)$.

(The corner-pair labels A, B, C match those of the May-13 night $(5, 2)$ j -formula paper; in the eighth-pass $(4, 3)$ notation these would be $\mu_{13}, \mu_{23}, \mu_{12}$, respectively.)

Same-row piece. A same-row $+q$ -piece exists for T_{n-1} when there is a removable corner c and an SYT of λ with $n-1$ at the cell immediately left of c (same row) and n at c . The candidates:

- At $c_1 = (1, 5)$: requires $T(1, 4) = n-1$, $T(1, 5) = n$. Columns 4 and 5 contain only cells $(1, 4)$ and $(1, 5)$ respectively (since $\lambda_2 = 2$), so no column-strict constraint below them. The remaining $n-2$ entries fill $\lambda \setminus \{(1, 4), (1, 5)\}$, which is the shape $\mu_h := (3, 2, 1^q)$, a valid partition. *Same-row piece at c_1 with inner shape μ_h .*
- At $c_2 = (2, 2)$: requires $T(2, 1) = n-1$, $T(2, 2) = n$. But then column-1 strict increase forces $T(3, 1) > T(2, 1) = n-1$, so $T(3, 1) = n$; conflict with $T(2, 2) = n$ (at $q \geq 1$). *No same-row piece at c_2 .*
- At $c_3 = (q+2, 1)$: no cell to the left of column 1. *No same-row piece at c_3 .*

No other $+q$ -pieces. Same-column pairs at column 1 give -1 -eigenvectors of T_{n-1} (not in E_{n-1}^+).

Lemma 10 (Four-piece decomposition of E_{n-1}^+). *Define embeddings $\Phi_X : V_{\mu_X} \hookrightarrow V_\lambda$ for $X \in \{A, B, C\}$ on the seminormal basis by*

$$\Phi_X(v_{T'}) := v_{T_+^X} + \tau_X v_{T_-^X},$$

where T_+^X, T_-^X are the two SYTs of λ obtained from T' by placing $\{n-1, n\}$ at the corresponding corner pair in the two possible orders, and $\tau_X \in \mathbb{Q}(q)^\times$ is the Hoefsmit 2-block coefficient at the corresponding axial distance. Define $\Phi_h : V_{\mu_h} \hookrightarrow V_\lambda$ on the seminormal basis by $\Phi_h(v_{T''}) := v_{T'''}$, where T''' is the unique SYT of λ obtained from T'' by placing $n-1$ at $(1, 4)$ and n at $(1, 5)$. Then:

- (i) Each Φ_X ($X \in \{A, B, C, h\}$) is injective and T_i -equivariant for $1 \leq i \leq n-3$.
- (ii) The four images in V_λ have pairwise disjoint seminormal supports and satisfy

$$R'_n V_\lambda = E_{n-1}^+|_{V_\lambda} = \Phi_A(V_{\mu_A}) \oplus \Phi_B(V_{\mu_B}) \oplus \Phi_C(V_{\mu_C}) \oplus \Phi_h(V_{\mu_h}).$$

Proof. (i) Injectivity and T_i -equivariance for $i \leq n-3$ are standard (May-20 Lemma 3.2; May-14 seventh pass Lemma 3.2): for $i \leq n-3$, the swap s_i moves only entries $i, i+1 \leq n-2$, both inside the “inner” cells of μ_X (or μ_h); Hoefsmit coefficients depend only on inner cells, so T_i commutes with the corresponding Φ .

(ii) Pairwise disjointness: each Φ ’s image is supported on SYTs of λ where $\{n-1, n\}$ occupies a specific *unordered* pair of cells:

- Φ_A : $\{(1, 5), (q+2, 1)\}$.
- Φ_B : $\{(2, 2), (q+2, 1)\}$.
- Φ_C : $\{(1, 5), (2, 2)\}$.
- Φ_h : $\{(1, 4), (1, 5)\}$ (specifically $n-1$ at $(1, 4)$, n at $(1, 5)$).

The four unordered cell pairs are pairwise distinct, so the supports are pairwise disjoint.

Dimension match: $\dim \Phi_X(V_{\mu_X}) = f^{\mu_X}$ (injective), and the $+q$ -eigenvectors of T_{n-1} in V_λ partition into the above four classes by where $\{n-1, n\}$ sit. By Theorem 8, $R'_n V_\lambda = E_{n-1}^+|_{V_\lambda}$, so the direct-sum decomposition follows.

[Computationally verified at $q \in \{3, 4, 5\}$: $\dim R'_n V_\lambda$ equals $f^{\mu_A} + f^{\mu_B} + f^{\mu_C} + f^{\mu_h}$. See [~/projects/scratch/2026-05-14-extended-sign-kill-52/verify_decomp.py](#).] $\square$ $\square$

Remark 11. For $\hat{\lambda} = (5, 2)$, two of the four inner shapes are hooks ($\mu_B = (5, 1^q)$ and $\mu_C = (4, 1^{q+1})$). Compare with $\hat{\lambda} = (4, 3)$ (eighth pass), where only one of the four was a hook ($\mu_h = (4, 1^{q+1})$). The shape-dependent feature is: in $(5, 2)$, $\lambda_2 = 2$ is small enough that the same-row piece at c_1 strips two cells from row 1 leaving $(3, 2, 1^q)$ (non-hook). In $(4, 3)$, $\lambda_2 = 3$ is large enough that the analogous same-row strip at c_1 would force $T(2, 3) > n-1$, killing that branch — but the same-row at $c_2 = (2, 3)$ strips two cells from row 2 leaving $(4, 1^{q+1})$ (hook). The total piece count is the same (four), but the hook count flips.

4 The two-branch reduction

Theorem 12 (Two-branch reduction). *For $\lambda = (5, 2, 1^q)$ with $q \geq 3$,*

$$B = (T_{n-5}+1)(T_{n-4}+1)(T_{n-3}+1)(T_{n-2}+1) \left[\Phi_A(B_{\ell(\mu_A)+1}^{(\mu_A)} V_{\mu_A}) + \Phi_B(B_{\ell(\mu_B)+1}^{(\mu_B)} V_{\mu_B}) \right]. \quad (1)$$

Proof. We follow the May-15/May-20 pull-through, applied separately to each of the four pieces of $R'_n V_\lambda$ from Lemma 10, then show two pieces vanish.

Step 1 (pulling four R' -blocks through Φ). Let $\Phi \in \{\Phi_A, \Phi_B, \Phi_C, \Phi_h\}$ and μ the corresponding inner shape.

Decompose $R'_{n-1} = (T_{n-2}+1)U_1$ where $U_1 = (T_{n-3}+1)\cdots(T_1+1) = R'_{n-2}^{(\mu)} \in H_q(S_{n-2})$. Since $T_1, \dots, T_{n-3}$ commute with T_{n-1} (index gaps ≥ 2), U_1 preserves $E_{n-1}^+|_{V_\lambda}$. By Lemma 10(i), $U_1\Phi(v) = \Phi(R'_{n-2}^{(\mu)}v)$ for $v \in V_\mu$. Hence

$$R'_{n-1}\Phi(V_\mu) = (T_{n-2}+1)\Phi(R'_{n-2}^{(\mu)}V_\mu).$$

Iterating the same decomposition for $R'_{n-2}, R'_{n-3}, R'_{n-4}$, using that each pulled-out T_j factor commutes with the next inner R' (index gaps ≥ 2):

$$R'_{n-4}R'_{n-3}R'_{n-2}R'_{n-1}\Phi(V_\mu) = (T_{n-5}+1)(T_{n-4}+1)(T_{n-3}+1)(T_{n-2}+1)\Phi(Q_\mu V_\mu), \quad (2)$$

where

$$Q_\mu := R'_{n-5}^{(\mu)} R'_{n-4}^{(\mu)} R'_{n-3}^{(\mu)} R'_{n-2}^{(\mu)}.$$

Step 2 (combining and applying Phase A). By Lemma 10(ii) and Theorem 8,

$$\begin{aligned} B &= \Omega V_\lambda = R'_{n-4} R'_{n-3} R'_{n-2} R'_{n-1} R'_n V_\lambda \\ &= R'_{n-4} R'_{n-3} R'_{n-2} R'_{n-1} \bigoplus_{X \in \{A, B, C, h\}} \Phi_X(V_{\mu_X}) \\ &= (T_{n-5}+1)(T_{n-4}+1)(T_{n-3}+1)(T_{n-2}+1) \sum_X \Phi_X(I_X), \end{aligned}$$

where $I_X := Q_{\mu_X} V_{\mu_X}$.

Step 3 (the μ_C piece vanishes). $\mu_C = (4, 1^{q+1})$ has $\ell(\mu_C) = q+2 = n-5$, hence $\ell(\mu_C)+1 = n-4$. The inner quadruple product is

$$I_C = R'_{n-5}^{(\mu_C)} (R'_{n-4}^{(\mu_C)} R'_{n-3}^{(\mu_C)} R'_{n-2}^{(\mu_C)} V_{\mu_C}) = R'_{n-5}^{(\mu_C)} B_{n-4}^{(\mu_C)} V_{\mu_C}.$$

By Theorem 6 applied to $\mu_C = (4, 1^{q+1})$ with $k = 4$, $m = q+1$ (hypothesis $m \geq \max(k, 2) = 4$, i.e., $q \geq 3$ ✓), $B_{n-4}^{(\mu_C)} V_{\mu_C} \subseteq E_1^-|_{V_{\mu_C}}$. The rightmost factor of $R'_{n-5}^{(\mu_C)}$ is (T_1+1) , which annihilates $E_1^-|_{V_{\mu_C}}$. Hence $I_C = 0$.

Step 4 (the μ_h piece vanishes). $\mu_h = (3, 2, 1^q)$ has $\ell(\mu_h) = q+2 = n-5$, hence $\ell(\mu_h)+1 = n-4$. Same argument as Step 3:

$$I_h = R'_{n-5}^{(\mu_h)} B_{n-4}^{(\mu_h)} V_{\mu_h}.$$

By Theorem 3 applied to $\mu_h = (3, 2, 1^q)$ with $q' = q \geq 3$ (joint hypothesis of the theorem and the present proof; we will tighten to $q \geq 5$ below), $B_{n-4}^{(\mu_h)} V_{\mu_h} \subseteq E_1^-|_{V_{\mu_h}}$ (the $j = 1$ instance of Theorem 3's chain). The rightmost (T_1+1) of $R'_{n-5}^{(\mu_h)}$ kills this, so $I_h = 0$.

Step 5 (the μ_A and μ_B pieces are standard $B^{(\mu_X)}$). Both $\mu_A = (4, 2, 1^{q-1})$ and $\mu_B = (5, 1^q)$ have $\ell(\mu_X) = q+1 = n-6$, hence $\ell(\mu_X)+1 = n-5$. The inner quadruple product runs from $R'_{n-5}^{(\mu_X)}$ to $R'_{n-2}^{(\mu_X)}$, exactly the four blocks of $\Omega^{(\mu_X)}$. Therefore

$$I_A = B_{\ell(\mu_A)+1}^{(\mu_A)} V_{\mu_A}, \quad I_B = B_{\ell(\mu_B)+1}^{(\mu_B)} V_{\mu_B}.$$

Combining Steps 2–5 gives (1). □

Remark 13. The dimensions match:

$$\dim B = f^{(4,1)} = 4, \quad \dim B_{\ell+1}^{(\mu_A)} = f^{(3,1)} = 3, \quad \dim B_{\ell+1}^{(\mu_B)} = f^{(4)} = 1,$$

and $4 = 3 + 1$ confirms that both contributing Phi-pieces inject and the four outer (T_j+1) factors are jointly injective on the sum $\Phi_A(I_A) + \Phi_B(I_B)$ (computationally verified at $q = 5$).

5 Main theorem

Theorem 14 (Main). For $\lambda = (5, 2, 1^q) \vdash n = q+7$ with $q \geq 5$,

$$B \subseteq \bigcap_{j=1}^{q-4} E_j^-|_{V_\lambda}.$$

Proof. By Remark 2, it suffices to show that every $T \in \text{supp}(B)$ satisfies $p(T) \geq q - 3 = n - 10$. We use the two-branch reduction (1) together with prefix bounds on the two inner $B_{\ell+1}^{(\mu_X)}$ pieces and Lemma 9 on the four outer factors.

Step 1 (prefix bound on the Φ_A -image). $\mu_A = (4, 2, 1^{q-1})$ has size $n - 2$, $\ell(\mu_A) = q + 1$, and trailing-1-count $q' = q - 1$. Since $q \geq 5$ we have $q' \geq 4$, so Theorem 4 applies, giving

$$p(T') \geq q' - 2 = q - 3 \quad \text{for every } T' \in \text{supp}(B_{\ell(\mu_A)+1}^{(\mu_A)} V_{\mu_A}).$$

The Phi-embedding Φ_A adds $\{n - 1, n\}$ at $\{c_1, c_3\} = \{(1, 5), (q + 2, 1)\}$. The added cells: $(1, 5)$ sits in column 5 (not column 1); $(q + 2, 1)$ sits at row $q + 2 = n - 5 > q + 1 = \ell(\mu_A)$, i.e., *below* all column-1 cells of μ_A . So column-1 rows $1, \dots, q + 1$ of any extension SYT $T_{\pm}^A$ match the column-1 entries of T' , and $p(T_{\pm}^A) \geq p(T') \geq q - 3$.

Step 2 (prefix bound on the Φ_B -image). $\mu_B = (5, 1^q)$, a hook with $k = 5$, $m = q$. Since $q \geq 5$ we have $m \geq k = 5$, so Theorem 7 applies and gives

$$p(T') \geq m - k + 2 = q - 3 \quad \text{for every } T' \in \text{supp}(B_{\ell(\mu_B)+1}^{(\mu_B)} V_{\mu_B}).$$

The Phi-embedding Φ_B adds $\{n - 1, n\}$ at $\{c_2, c_3\} = \{(2, 2), (q + 2, 1)\}$. Cell $(2, 2)$ is column 2 (not column 1); cell $(q + 2, 1)$ is row $q + 2 = n - 5 > q + 1 = \ell(\mu_B)$, below μ_B 's column-1 cells. So by the same argument as Step 1, $p(T_{\pm}^B) \geq p(T') \geq q - 3$.

Step 3 (outer factors preserve the prefix). The four outer factors in (1) are $(T_j + 1)$ for $j \in \{n - 5, n - 4, n - 3, n - 2\}$. For prefix bound $P = q - 3 = n - 10$, the hypothesis $j \geq P + 1 = q - 2 = n - 9$ of Lemma 9 requires $j \geq n - 9$. The smallest index $j = n - 5$ satisfies $n - 5 \geq n - 9$ iff $9 \geq 5$ ✓, true unconditionally. Applying Lemma 9 four times in sequence, the prefix bound $\geq q - 3$ is preserved.

Conclusion. By Steps 1–3 and (1), every $T \in \text{supp}(B)$ has $p(T) \geq q - 3$. By Remark 2, $B \subseteq E_j^-$ for $1 \leq j \leq q - 4 = \tau(\lambda)$. $\square$

Remark 15 (Hypotheses). The hypothesis $q \geq 5$ is the joint requirement of:

- Theorem 4 at $q' = q - 1 \geq 4$, i.e., $q \geq 5$ (for μ_A).
- Theorem 7 at $\mu_B = (5, 1^q)$ with $k = 5$, $m = q \geq 5$ (for μ_B).
- Theorem 3 at $\mu_h = (3, 2, 1^q)$ with $q' = q \geq 3$ (for μ_h vanishing in Step 4 of Theorem 12).
- Theorem 6 at $\mu_C = (4, 1^{q+1})$ with $k = 4$, $m = q + 1 \geq 4$, i.e., $q \geq 3$ (for μ_C vanishing in Step 3).

The threshold for the two contributing-branch theorems ($q \geq 5$) is tight: at $q = 4$ the prediction $\tau = q - 4 = 0$ is vacuous, and at $q \leq 3$ the prediction is negative. Thus the hypothesis $q \geq 5$ matches the first q at which $\tau \geq 1$.

6 Sharpness (computational)

The predicted threshold $\tau(\hat{\lambda}) = q + 3 - \hat{\lambda}_1 - \hat{\lambda}_2 = q - 4$ asserts both $B \subseteq \bigcap_{j=1}^{q-4} E_j^-$ (Theorem 14) and $B \not\subseteq E_{q-3}^-$. We verify the latter computationally for the first non-vacuous case $q = 5$ (where $q - 3 = 2$).

Theorem 16 (Sharpness at $q = 5$, computational). *For $\lambda = (5, 2, 1^5)$ at $n = 12$, there exists $T \in \text{supp}(B)$ with $p(T) = 2 = q - 3$ exactly. Consequently $B \not\subseteq E_{q-3}^-|_{V_\lambda} = E_2^-|_{V_\lambda}$.*

Computational evidence. Applying Ω to the first 40 Hoefsmit basis vectors and recording the support (Section 7) gives a probe set in $\text{supp}(B)$ of size 524, with prefix histogram $\{2 : 168, 3 : 168, 4 : 112, 5 : 56, 6 : 16, 7 : 4\}$. An explicit witness at $p(T) = 2$:

$$T(1, \cdot) = (1, 3, 4, 5, 6), \quad T(2, \cdot) = (2, 7), \quad T(3..7, 1) = (8, 9, 10, 11, 12).$$

Column-1 entries are $(1, 2, 8, 9, 10, 11, 12)$, so $p(T) = 2$ exactly. Hence $T \in \text{supp}(B)$ has $p(T) < q-2$, and by the contrapositive of Remark 2 (applied at $j = q - 3 = 2$), $B \not\subseteq E_2^-$. $\square$ $\square$

Remark 17. A structural-sharpness argument analogous to the May-14 sixth-pass $(4, 2)$ Lemma 5.1 should locate entry $n - 8$ at $(1, 3)$ in a forced position and produce an off-diagonal T_{q-3} -swap to a tableau violating the prefix bound, contradicting $B \subseteq E_{q-3}^-$. We omit the detailed write-up of this for the present paper; the mechanism is identical to the $(4, 2)$ proof.

7 Computational verification

We verify Theorem 14 and the structural decomposition at $q = 5$ (so $n = 12$, $\dim V_\lambda = 1728$).

Decomposition. At $q = 5$ ($n = 12$, $\dim V_\lambda = 1728$),

$$\begin{aligned} f^{\mu_A} &= f^{(4,2,1^4)} = 350, & f^{\mu_B} &= f^{(5,1^5)} = 126, \\ f^{\mu_C} &= f^{(4,1^6)} = 84, & f^{\mu_h} &= f^{(3,2,1^5)} = 160. \end{aligned}$$

Sum = 720 = $\dim R'_n V_\lambda$, matching Lemma 10. (Also verified at $q = 3$: $90 + 35 + 35 + 64 = 224$; $q = 4$: $189 + 70 + 56 + 105 = 420$.)

E_1^- containment at $q = 5$. Apply Ω to five independent random vectors in $V_\lambda \pmod{P = 100\,003}$ at $q = 11$) and check $(T_1 + 1)\Omega v = 0$ on each. All five tests pass; each Ωv has 524 nonzero entries, and each $(T_1 + 1)\Omega v$ is the zero vector. Probing $\dim B$ via Ω applied to the first 40 basis vectors gives rank 4, matching $f^{(4,1)} = 4$.

Prefix bound and sharpness witness. The same probe (see Theorem 16) gives prefix histogram $\{2, 3, 4, 5, 6, 7\}$ with $\min = 2 = q - 3$, matching the predicted threshold and confirming sharpness at $q = 5$.

Scripts: `~/projects/scratch/2026-05-14-extended-sign-kill-52/verify_decomp.py`, `verify_q5_sparse`

8 Updated table of proved $\widehat{\lambda}$ rows

9 Discussion

9.1 The $\widehat{\lambda}_1 + \widehat{\lambda}_2 = 7$ row is now complete

The eighth-pass $(4, 3)$ paper proved the first member of the $\widehat{\lambda}_1 + \widehat{\lambda}_2 = 7$ row. This paper proves the second and last member, $(5, 2)$. The two share predicted threshold $\tau = q - 4$ and the same hypothesis $q \geq 5$. Combined with $(4, 4)$ at $\tau = q - 5$ (ninth pass), the $|\widehat{\lambda}| \leq 8$ stratum is closed except for $(5, 3)$ at $|\widehat{\lambda}| = 8$.

Table 1: Status of the extended sign-kill conjecture by $\widehat{\lambda}$. The τ column lists the predicted threshold $\tau(\widehat{\lambda}) = q + 3 - \widehat{\lambda}_1 - \widehat{\lambda}_2$.

$\widehat{\lambda}$	predicted τ	status	paper
(2, 2)	$q - 1$	proved $q \geq 2$	May-14 fourth pass
(3, 2)	$q - 2$	proved $q \geq 3$	May-14 fifth pass
(4, 2)	$q - 3$	proved $q \geq 4$ (gap closable)	May-14 sixth pass
(3, 3)	$q - 3$	proved $q \geq 4$	May-14 seventh pass
(4, 3)	$q - 4$	proved $q \geq 5$	May-14 eighth pass
(4, 4)	$q - 5$	proved $q \geq 6$	May-14 ninth pass
(5, 2)	$q - 4$	proved $q \geq 5$ (this paper)	May-14 tenth pass

9.2 Hooks as contributing branches

In $\widehat{\lambda} = (5, 2)$, the contributing branch $\mu_B = (5, 1^q)$ is a hook. This is the first proof in the May-14 series where a hook gives a non-zero contribution to B at the $\ell + 1$ index (rather than vanishing as μ_C does). The mechanism: the hook μ_B has $\ell(\mu_B) = q + 1 = n - 6$, so the inner $\Omega^{(\mu_B)}$ has four blocks matching the four “standard-position” blocks of the two-branch reduction, and the hook support lemma at $k = 5$, $m = q$ gives a clean prefix bound $\geq q - 3$. The contribution $B_{\ell+1}^{(\mu_B)} V_{\mu_B}$ is one-dimensional (hook dim), so Φ_B contributes exactly 1 to $\dim B$.

9.3 Toward $\widehat{\lambda} = (5, 3)$

$\lambda = (5, 3, 1^q)$ has corners $c_1 = (1, 5)$, $c_2 = (2, 3)$, $c_3 = (q + 2, 1)$. Three corner pairs and (possibly) one same-row piece at c_2 give the same four-piece structure. Inner shapes for the corner pairs:

- $\mu_{12} = (4, 2, 1^q) — \widehat{\lambda} = (4, 2)$ shape.
- $\mu_{13} = (4, 3, 1^{q-1}) — \widehat{\lambda} = (4, 3)$ shape, recursive on eighth pass.
- $\mu_{23} = (5, 2, 1^{q-1}) — \widehat{\lambda} = (5, 2)$ shape, recursive on *this* paper.

The same-row piece at $c_2 = (2, 3)$ would have inner shape $\lambda \setminus \{(2, 2), (2, 3)\} = (5, 1^{q+1})$, a hook. (The candidate same-row at $c_1 = (1, 5)$ would have inner shape $(3, 3, 1^q)$, which requires the column-4 constraint $T(1, 4) < T(2, 4)$ to not bite; cell $(2, 4)$ does not exist in $(5, 3, 1^q)$ since $\lambda_2 = 3$, so there is no constraint and the candidate *does* contribute. So $\widehat{\lambda} = (5, 3)$ may actually have *two* same-row pieces, giving five total pieces in E_{n-1}^+ .) Investigation of $(5, 3)$ is left for a follow-up session.

9.4 Push status

71 unpushed commits on `clio-vega/proofs` (after today’s tenth-pass commit, assuming the count from the ninth-pass paper held at 70 before this). PAT issue from May 6 persists — please refresh.

10 Status

Established.

- Lemma 10: four-piece E_{n-1}^+ decomposition for $\lambda = (5, 2, 1^q)$ at $q \geq 3$.

- Theorem 12: two-branch reduction (two contributing, two vanishing) at $q \geq 3$.
- Theorem 14: $B \subseteq E_j^-$ for $j \leq q - 4$ at $q \geq 5$.
- Theorem 16: sharpness at $j = q - 3$ for $q = 5$ (computational witness).

Open.

- Structural (non-computational) sharpness across all $q \geq 5$ (analogous to the sixth-pass Lemma 5.1; mechanism stated in Remark 17).
- Sharpness witnesses at $q \in \{6, 7, \dots\}$ (consistent computationally up to memory limits; not run in this session).
- The next family $\widehat{\lambda} = (5, 3)$ (potentially five-piece decomposition; section 9).